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Sample Average Approximation

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- Motivation: computational complexity
- Sample average approximation
- Convergence and optimality gap
- Exercise



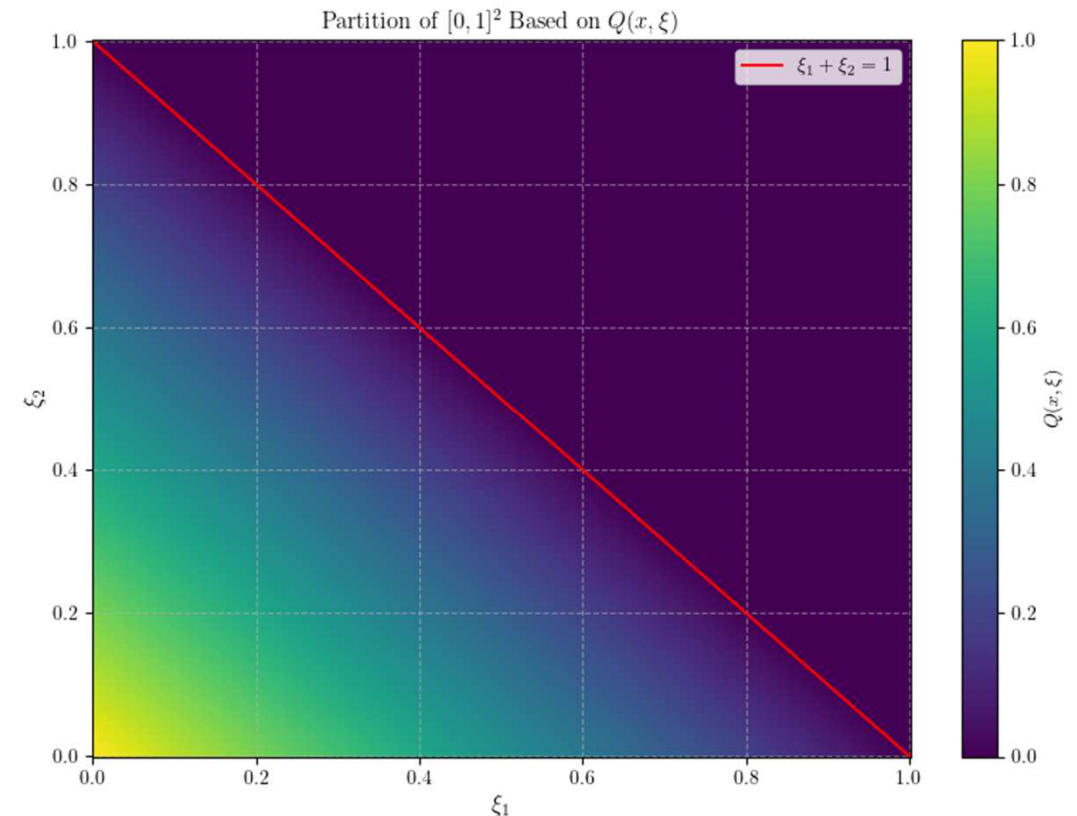
- A super simple example

$$Q(x, \xi) = \min_y \{y \mid y \geq 1 - (\xi_1 + \xi_2), y \geq 0\} \text{ with } \xi \sim U[0, 1]^2$$

$$\mathbb{E}[Q(x, \xi)] = \int_{[0,1]^2} Q(x, \xi) d\xi$$

- Challenges:

- Multidimensional ξ
- Q nonlinear in ξ
- Must partition the support set (and identify the feasible region)
- Integral over irregular region

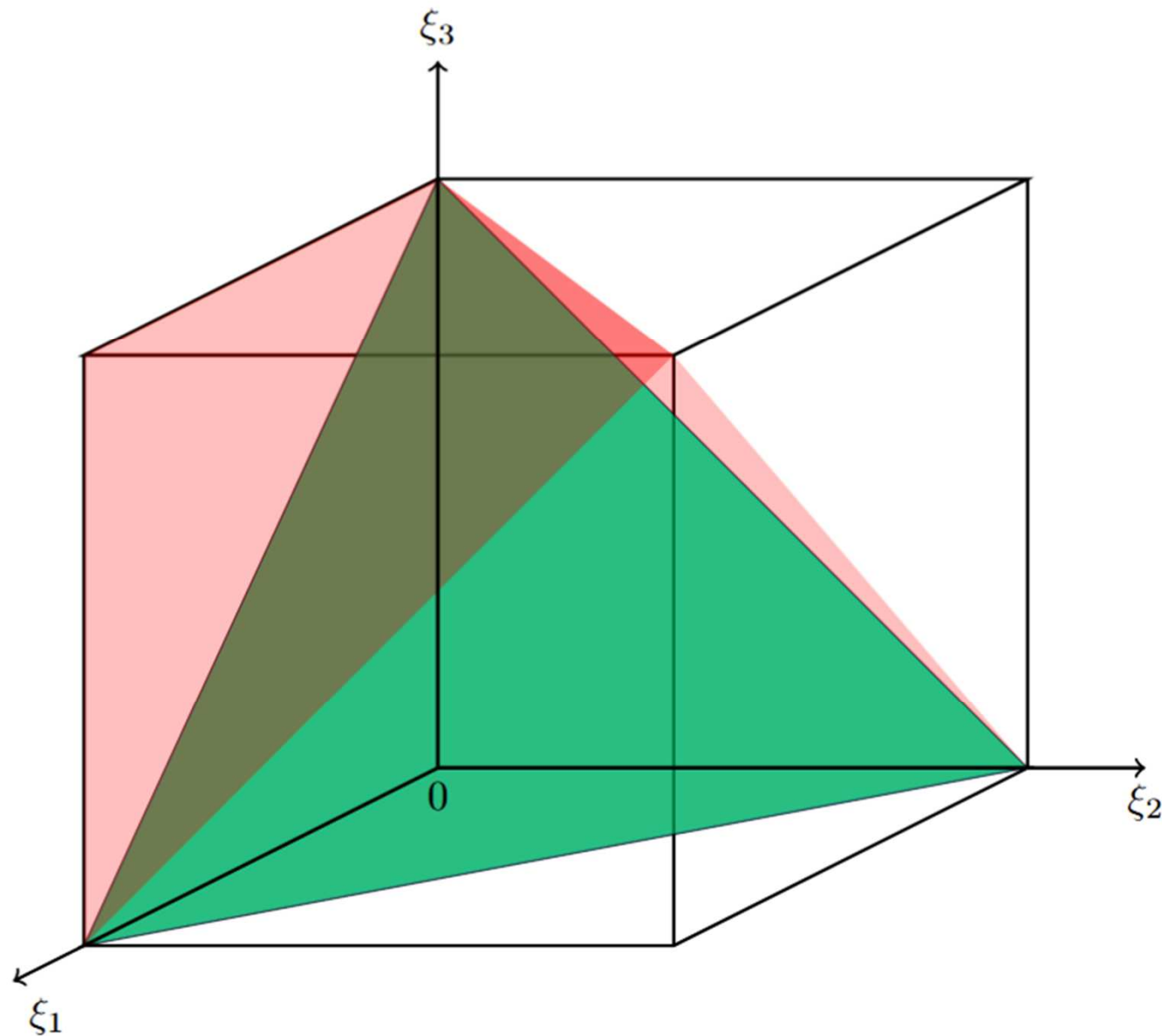




Solving SP problems



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- **Complexity result:**

Proposition (Dyer and Stougie 2006) Calculating the expected recourse value $\mathbb{E}[Q(x, \xi)]$ for a fixed first-stage decision x is $\#\mathbf{P}$ -hard even if the two-stage SP exhibits fixed recourse and the random vector ξ is governed by a uniform distribution supported on $[0, 1]^K$

$$Q(x, \xi) = \min_y \{ c^\top y \mid \overbrace{W y = h(\xi) - T(\xi)x}^{\text{coupling between } \xi \text{ and } y}, y \geq 0 \}$$

$$\mathbb{E}[Q(x, \xi)] = \int_{[0,1]^K} Q(x, \xi) d\xi$$



- Analytical solution in special cases, when expectations can be found analytically (**quite rare** in practice)
- **General case:** stochastic sampling strategy
 - **Sample average approximation (SAA)** and solved using correspond deterministic opt. techniques
 - **Stochastic sub(quasi)-gradient methods** for convex problems



$$\begin{cases} \min & L(x) = c^\top x + \mathbb{E}[Q(x, \xi)] \\ \text{s.t.} & x \in \mathcal{X} \end{cases}$$

- **Idea:** replace the expectation with **a sampled model** when the support Ξ is continuous
- Generate N i.i.d samples $\xi_1, \xi_2, \dots, \xi_N$, each with $p_i = 1/N$
- Approximate the expected recourse function

$$\begin{cases} \min & L_N = c^\top x + \frac{1}{N} \sum_{i=1}^N Q(x, \xi_i) \\ \text{s.t.} & x \in \mathcal{X} \end{cases}$$

- Now a **deterministic** SAA problem

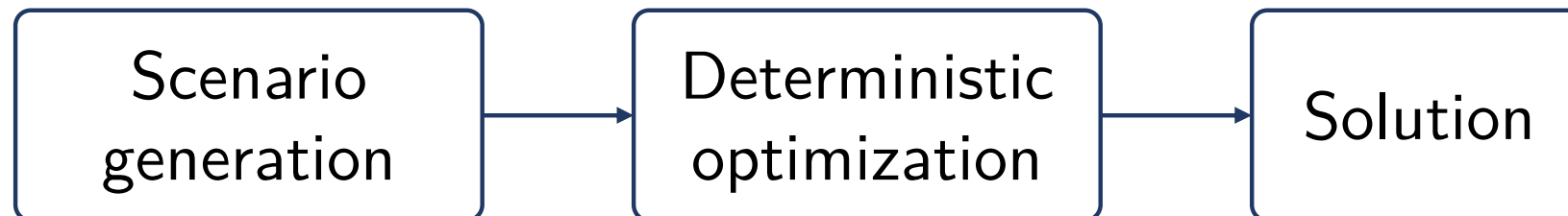
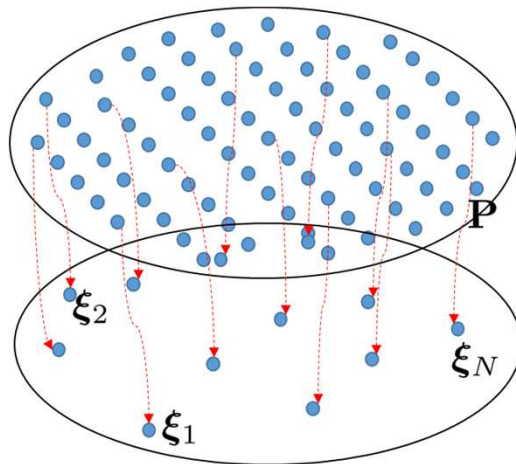


Example



- The production planning example, SAA deterministic counterpart:

$$\begin{aligned} \min \quad & 2x_1 + 3x_2 + \sum_{i=1}^N p_i [7y_1(\xi_i) + 12y_2(\xi_i)] \\ \text{s.t.} \quad & x_1 + x_2 \leq 100 \\ & x_1, x_2 \geq 0 \\ & \left. \begin{aligned} \alpha(\xi_i)x_1 + 6x_2 + y_1(\xi_i) &\geq h_1(\xi_i) \\ 3x_1 + \beta(\xi_i)x_2 + y_2(\xi_i) &\geq h_2(\xi_i) \\ y_1(\xi_i), y_2(\xi_i) &\geq 0 \end{aligned} \right\} i = 1, \dots, N \end{aligned}$$



- Scenario generation: commonly use Monte Carlo sampling
- Often objective is to reduce scenario set to **manageable size**



- SAA is **NOT** a solution algorithm or technique
- One must still solve the problem using appropriate computational method
- Obtained SAA problem is **less hard** to solve than the original problem
- The complexity of the of the problem depends on the **variance of $Q(x, \xi)$** , not the number of scenarios



- Following conditions ensure good approximation of the SAA (Shapiro and Philpott, 2007):
 - i. it is possible to **generate a sample of realizations** of the random vector ξ
 - ii. for moderate values of the sample size, it is possible to **efficiently solve** the obtained SAA problem
 - iii. the true problem has **relatively complete recourse**
 - iv. **variability of $Q(x, \xi)$** is not “too large”



- Denote by L_N^* the optimal value of the SAA problem and $x_N^* \in O_N^*$ an optimal solution (set)
- How good is **the quality of the sampled SAA solution?**
- Does it approach the true optimal value L^* and the true set of optimal solution O^* (as N increases)?
- How fast?
- How to **validate the approximation** in practice?

$$\begin{array}{ccc} L_N^* & \xRightarrow{?} & L^* \\ \uparrow & & \uparrow \\ x_N^* \in O_N^* & & x^* \in O^* \end{array} \quad L(x) = c^\top x + \mathcal{Q}(x)$$

- All uncertainty lies in the second stage expected recourse



- Let us fix a point $x' \in \mathcal{X}$, the estimator $L_N(x')$ is **unbiased**, and have **asymptotically normal distribution**

$$\sqrt{N}[L_N(x') - L(x')] \xrightarrow{\mathcal{D}} Y(x') \sim \mathcal{N}(0, \mathbf{Var}[Q(x', \xi)])$$

- Under mild technical assumptions (compactness of feasible solution, lower semi-continuity of costs, etc.), as $N \rightarrow \infty$

→ **Law of Large Number extension:**

$$L_N^* \rightarrow L^* \text{ almost surely, w.p. 1}$$

→ **Convergence of controls:**

$$\mathbf{D}(O_N^*, O^*) \rightarrow 0 \text{ almost surely, w.p. 1}$$



- Generically, the estimators of the minimum are **biased**

$$\mathbb{E}_N[L_N^*] \leq \mathbb{E}_{N+1}[L_{N+1}^*] \leq L^*$$

Proof sketch:

- Let $(\xi_j)_{j \in N}$ be a sequence of i.i.d copies of ξ
- We have, for every $x' \in \mathcal{X}$

$$L_N(x') := c^\top x' + \frac{1}{N} \sum_{j=1}^N Q(x', \xi_j) \geq \inf_{x \in \mathcal{X}} L_N(x)$$

- Take the expectation on both side yields

$$\mathbb{E}_N [L_N(x')] = L(x') \geq \mathbb{E}_N \left[\inf_{x \in \mathcal{X}} L_N(x) \right] = \mathbb{E}_N[L_N^*]$$

- Take the infimum over all $x' \in \mathcal{X}$ to obtain

$$L^* = \inf_{x' \in \mathcal{X}} L(x') \geq \mathbb{E}_N[L_N^*]$$



- We now show that the bias is monotonically decreasing, note that

$$\begin{aligned} L_x &= c^\top x + \mathcal{Q}(x) & \mathcal{Q}_{N+1}(x) &:= \frac{1}{N+1} \sum_{j=1}^{N+1} \left[\frac{1}{N} \sum_{k \neq j} Q(x, \xi_k) \right] \\ \mathbb{E}[L_{N+1}^*] &= \mathbb{E} \left[\inf_{x \in \mathcal{X}} L_{N+1}(x) \right] = \mathbb{E} \left\{ \inf_{x \in \mathcal{X}} \left[c^\top x + \frac{1}{N+1} \sum_{j=1}^{N+1} \frac{1}{N} \sum_{k \neq j} Q(x, \xi_k) \right] \right\} \\ &\geq \mathbb{E} \left\{ \frac{1}{N+1} \sum_{j=1}^{N+1} \inf_{x_j \in \mathcal{X}} \left[c^\top x + \frac{1}{N} \sum_{k \neq j} Q(x_j, \xi_k) \right] \right\} \\ &= \frac{1}{N+1} \sum_{j=1}^{N+1} \mathbb{E} \left\{ \inf_{x_j \in \mathcal{X}} \left[c^\top x + \frac{1}{N} \sum_{k \neq j} Q(x_j, \xi_k) \right] \right\} \\ &= \frac{1}{N+1} \sum_{j=1}^{N+1} \mathbb{E}[L_N^*] = \mathbb{E}[L_N^*] \end{aligned}$$



- Under mild technical assumptions, the following holds:

$$\sqrt{N}(L_N^* - L^*) \xrightarrow{\mathcal{D}} \inf_{x \in O^*} Y(x) \text{ where } Y(x) \sim \mathcal{N}(0, \mathbf{Var}[Q(x, \xi)])$$

- If $O^* = \{x^*\}$ (**single optimal solution**), then

$$\sqrt{N}(L_N^* - L^*) \xrightarrow{\mathcal{D}} Y(x^*) \sim \mathcal{N}(0, \mathbf{Var}[Q(x^*, \xi)])$$

- The **bias** is strictly negative, of order $o(N^{-1/2})$ (if O^* is a singleton)

$$\mathbb{E}[L_N^*] - L^* = \frac{1}{\sqrt{N}} \mathbb{E} \left[\inf_{x \in O^*} Y(x) \right] + o\left(\frac{1}{\sqrt{N}}\right)$$



- Let us fix a significance level $\alpha \in (0,1)$, and estimate the sample size N which is needed for

$$\mathbb{P}\{O_N^* \subset O_\epsilon^*\} \geq 1 - \alpha$$



$$O_\epsilon^* = \{x' \in \mathcal{X} : L(x') < +\infty, L(x') \leq L^* + \epsilon\}$$

- We require (Eq. 3.17, Shapiro, A. 2003):

$$N \geq \frac{12\sigma^2}{\epsilon^2} \left(d \log \frac{2DL_z}{\epsilon} - \log \alpha \right)$$



- How to evaluate SAA solution quality when we don't know the $\text{Var}[Q(x^*, \xi)]$ and the constants?
- For a minimization problem, given a **feasible** x' , we have

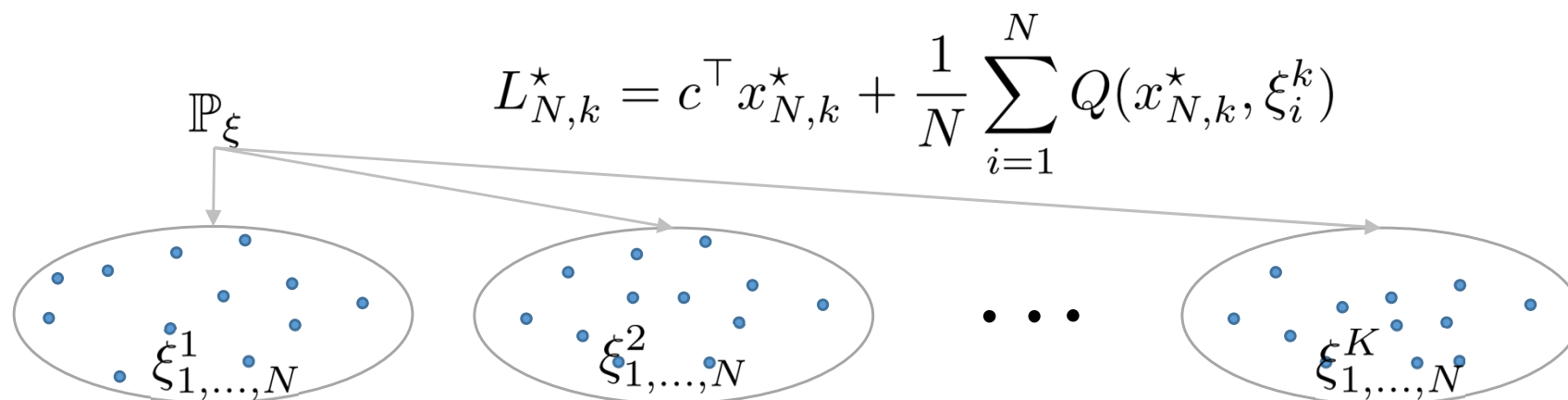
$$L(x') \geq L^*$$

- True optimality gap:

$$\text{gap}(x') = L(x') - L^*$$



- Since $\mathbb{E}_N[L_N^*] \leq \mathbb{E}_N[L_{N+1}^*] \leq L^*$
- $\mathbb{E}_N[L_N^*]$ can be estimated by **solving SAA problems K times independently**, obtaining estimator



$$\bar{L}_{N,K}^* = \frac{1}{K} \sum_{k=1}^K L_{N,k}^*$$

$$\hat{\sigma}_{N,K}^2 = \frac{1}{K-1} \sum_{k=1}^K (L_{N,k}^* - \bar{L}_{N,K}^*)^2$$



$$\bar{L}_{N,K}^* = \frac{1}{K} \sum_{k=1}^K L_{N,k}^*$$

Inverse CDF of Student's t -distribution $\nu = K - 1$ at $1 - \alpha$ (close to 1)

$$\text{LB}_{N,K} = \bar{L}_{N,K}^* - t_{\nu}^{-1}(1 - \alpha) \hat{\sigma}_{N,K}^2$$

$$\hat{\sigma}_{N,K}^2 = \frac{1}{K - 1} \sum_{k=1}^K (L_{N,k}^* - \bar{L}_{N,K}^*)^2$$

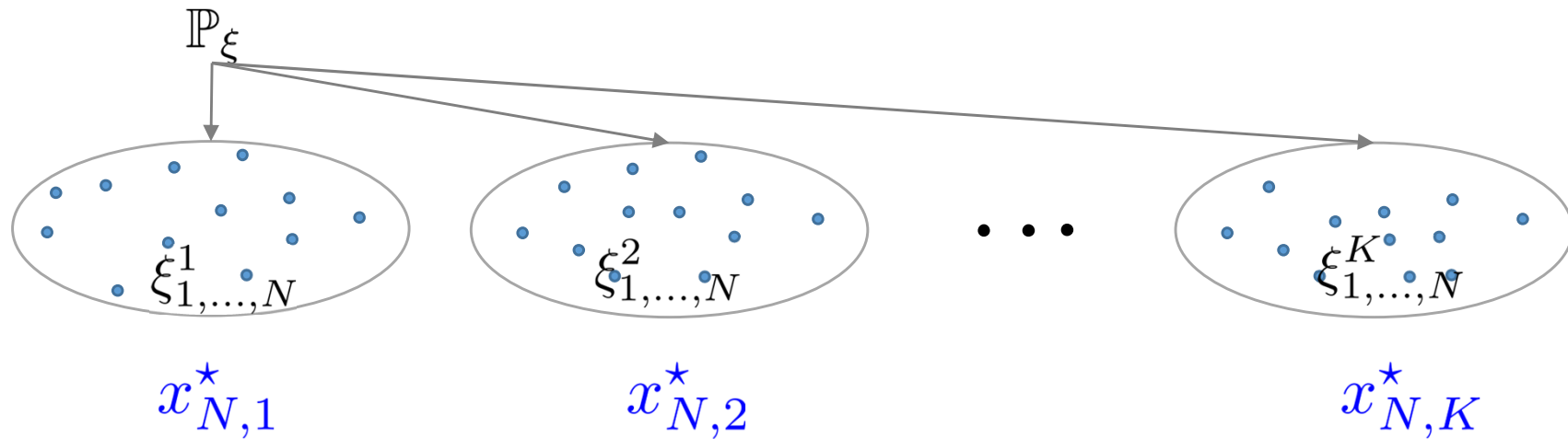
- Often sufficient to use **normal distribution**



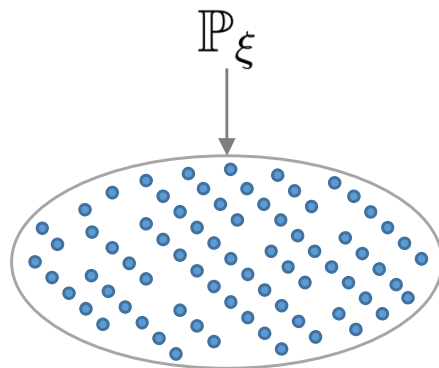
Upper bound



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$$L(x_{N,k}^*) \geq L^* \text{ if } x_{N,k}^* \text{ is feasible}$$



another independent validation dataset

$\xi_1^{\text{val}}, \xi_2^{\text{val}}, \dots, \xi_M^{\text{val}}$ with $M \gg N$

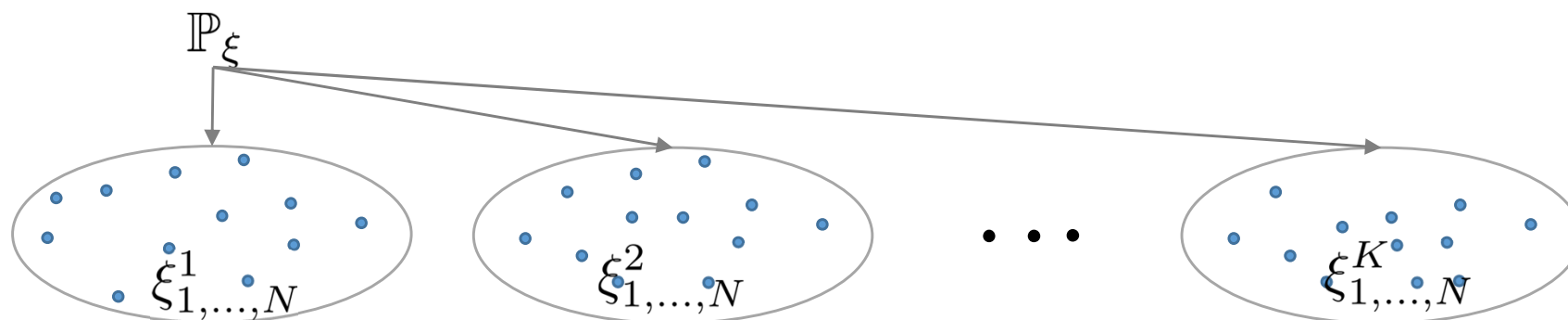
$$L(x_{N,k}^*) \approx L_M^{\text{val}}(x_{N,k}^*) = c^\top x_{N,k}^* + \frac{1}{M} \sum_{i=1}^M Q(x_{N,k}^*, \xi_i^{\text{val}})$$



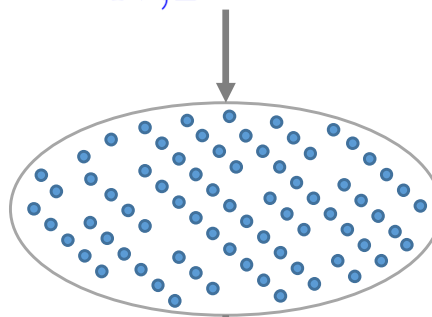
Upper bound



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$x_{N,1}^*$ $x_{N,2}^*$..., $x_{N,K}^*$



another **independent** validation dataset

$\xi_1^{\text{val}}, \xi_2^{\text{val}}, \dots, \xi_M^{\text{val}}$ with $M \gg N$

$L_M^{\text{val}}(x_{N,1}^*), L_M^{\text{val}}(x_{N,2}^*), \dots, L_M^{\text{val}}(x_{N,K}^*)$

The tightest UB = $\min_{k=1 \dots K} L_M^{\text{val}}(x_{N,k}^*)$



- Using CLT to construct the CI for $L_M^{\text{val}}(x_{N,k^\circ}^*)$ on different validation dataset M if needed

Inverse CDF of standard normal

$$\text{UB}(x_{N,k^\circ}^*) = \bar{L}_M^{\text{val}}(x_{N,k^\circ}^*) + \Phi^{-1}(1 - \alpha) \hat{\sigma}_M(x_{N,k^\circ}^*)$$

$$\hat{\sigma}_M^2(x_{N,k^\circ}^*) = \mathbf{Var}[L_M^{\text{val}}(x_{N,k^\circ}^*)]$$



- A statistically valid (with confidence at least $1 - 2\alpha$) bound on the true gap

$$\text{gap}(x_{N,k^\circ}^\star) = \text{UB}(x_{N,k^\circ}^\star) - \text{LB}_{N,K}$$

- The lower bound is **somewhat conservative** since the bias of the SAA estimator of the true optimal value could be significant for not “too large” sample size N .



How to choose N, K ?

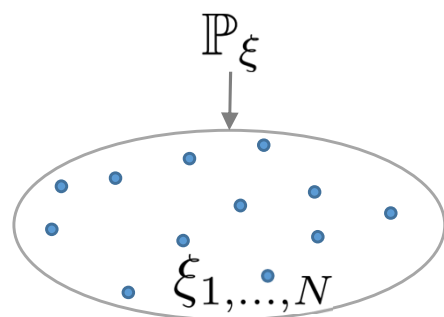


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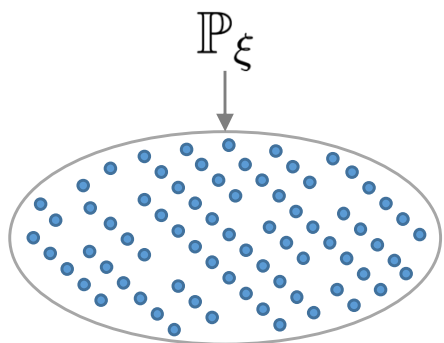
- The bigger the better, giving the smallest possible optimality gap
- If the computational complexity of solving the SAA problem increases **faster than linearly** in the sample size N , it may be more efficient to choose a smaller N and to generate and solve several SAA problems



- A practical way to **quickly check** if N is large enough



$$L_N^* = c^\top x_N^* + \frac{1}{N} \sum_{i=1}^N Q(x_N^*, \xi_i)$$



$\xi_{1\dots M}^{\text{val}}$ with $M \gg N$

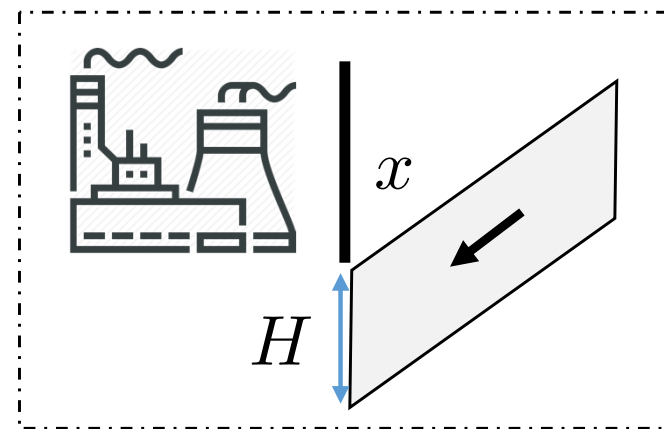
$$L_M^{\text{val}}(x_N^*) = c^\top x_N^* + \frac{1}{M} \sum_{i=1}^M Q(x_N^*, \xi_i^{\text{val}})$$

- If $L_M^{\text{val}}(x_N^*) \approx L_N^*$ our confidence that N is large enough is enhanced, **proceed to solving N -SSA K times**
- If not, increase N



Problem: protect a power plant from potential river flooding by building a dike with a height x meter

- Annualized investment cost $c \cdot x$
- **Uncertain** annual maximum flooding water height H
- Annual economic cost is linear related to the height of overflow $H - x$ when it's positive with **uncertain** coefficient b
- **Objective:** minimize investment cost + expected economic cost



$$\begin{aligned} \min_x \quad & cx + \mathbb{E}[b(H - x)_+] \\ \text{s.t.} \quad & x \geq 0 \end{aligned}$$



- $c = 0.9, H \sim \text{Gamma}(8, 4.5)$ and $b \sim \text{Lognorm}(0.7, 0.16^2)$

$$\begin{aligned} \min_x \quad & 0.9x + \frac{1}{N} \sum_{i=1}^N [b_i (H_i - x)_+] \\ \text{s.t.} \quad & 0 \leq x \leq 3 \end{aligned}$$

- We solve the SAA (LP problem) with $N = 20, 200, 2000$
- Validation set uses $M = 100\,000$



SAA example

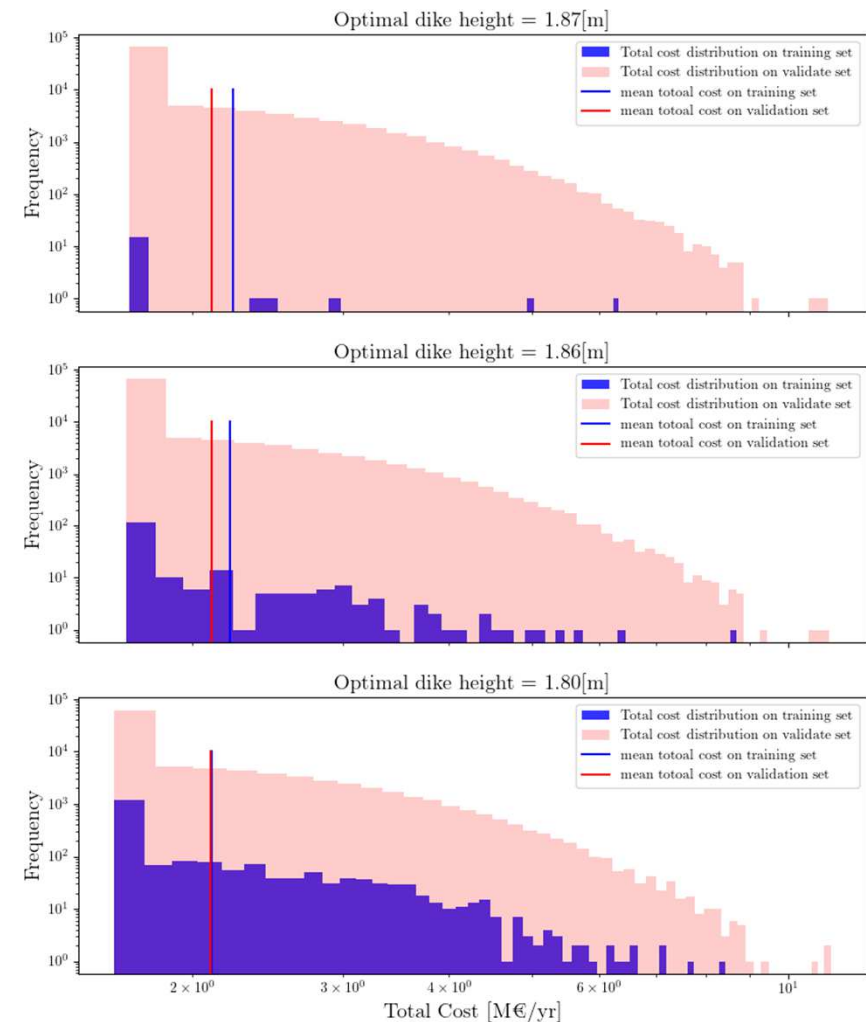


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N	20	200	2000
$x_N^\star$	1.87	1.86	1.80
$L_N^\star$	2.235	2.213	2.108
$L_M^{\text{val}}(x_N^\star)$	2.108	2.107	2.104

- $N = 2000$ is probably fine
- Proceed to compute the **optimality gap**, by solving $K = 20$ independent 2000-SAA instances:

$$\text{gap}(x_{N,k^\circ}^\star = 1.793) = 2.112 - 2.078$$

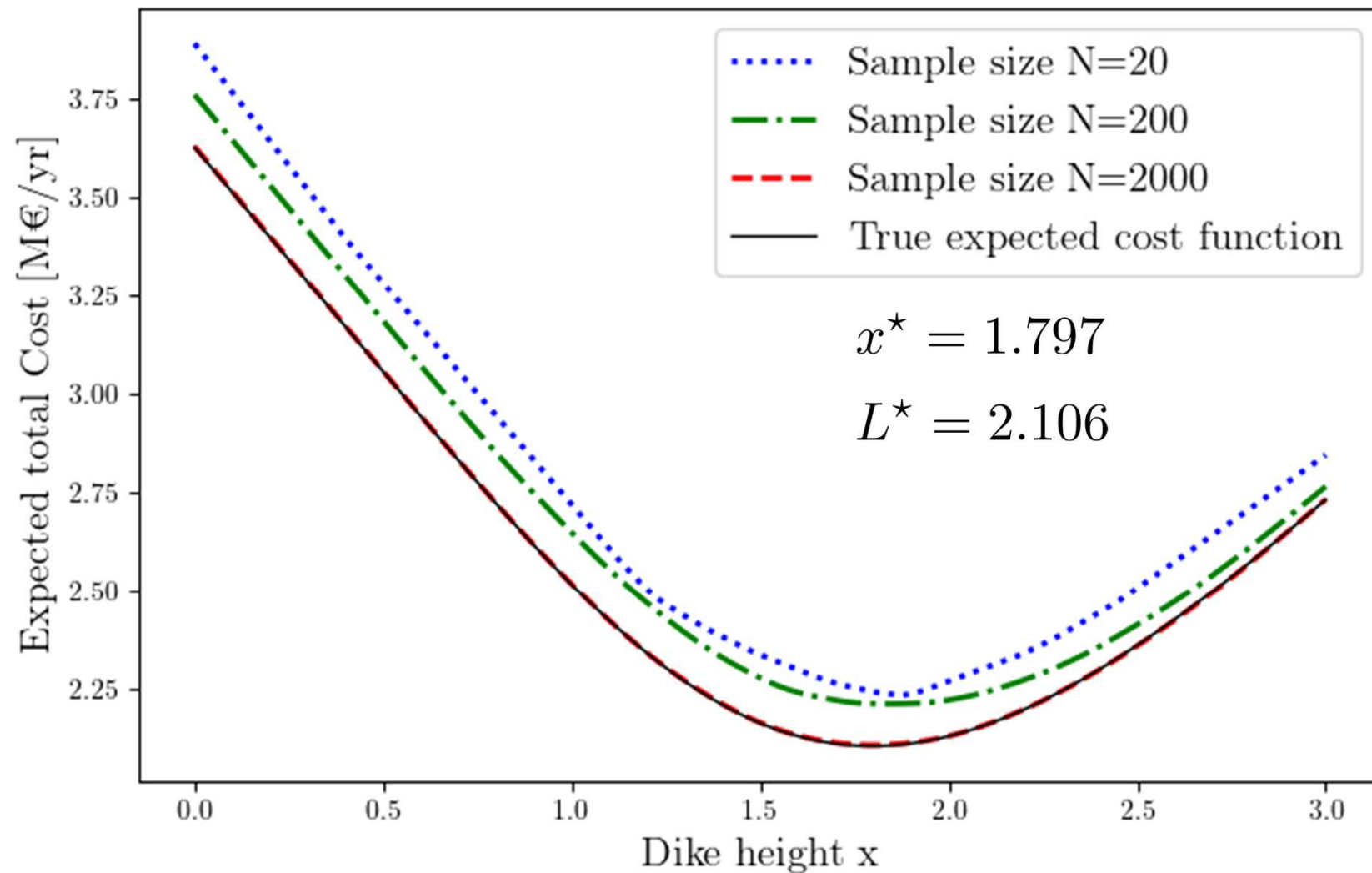




SAA example



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- (Only for this simple problem)

$$N \geq \frac{12\sigma^2}{(\epsilon = 0.1)^2} \left(d \log \frac{2DL}{\epsilon = 0.1} - \log(\alpha = 0.05) \right) \approx 7861$$

$$\hat{\sigma}^2 \approx 0.8$$

$$d = 1$$

$$D = 3$$

$$L \approx |0.9 - \max b| \approx 3$$



- Implement and solve the gasoline production planning recourse using SAA



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